Problem J
Stenographer

Time limit: 2 seconds

Musaddiq has loved typing for as long as he can remember. As a child he raced his cousins on the family computer, and at university he became the fastest typist in his class. Now he has set his heart on becoming a court stenographer, so every evening he practises on an old stenotype machine he found at a market in Peshawar.

The machine has one key for every lowercase Latin letter. Pressing the key of letter c takes w_c units of effort. The machine is old, and some of its keys are jammed: for a jammed key $w_c = -1$, and that key cannot be pressed at all.

Like every stenotype, the machine also understands *chords*. There are m chords, and the j-th chord is described by three letters z_j, x_j and y_j. Musaddiq may choose to use this chord by first producing the letter x_j and then producing the letter y_j. The chord replaces these two results with a single letter z_j, with no additional effort. Each of x_j and y_j may itself be produced by pressing its key or by using a chord, so chords can be nested to any finite depth. Each chord may be used any number of times. The effort spent on a letter is the total effort of all key presses used to produce it.

The letters of s are produced independently, in order. Chords are applied only when Musaddiq chooses to use them, so completed letters of s are not automatically merged.

Tonight Musaddiq wants to type the string s. Help him find the minimum total effort needed to type it, or tell him that it cannot be typed on this machine.

Input

The first line contains a single integer t ($1 \le t \le 10^4$), the number of test cases.

The first line of each test case contains 26 integers $w_a, w_b, \ldots, w_z$ ($w_c = -1$ or $1 \le w_c \le 10^6$), the effort of pressing each key, or -1 if the key is jammed.

The second line contains a single integer m ($0 \le m \le 2 \cdot 10^5$), the number of chords.

Each of the next m lines contains three lowercase Latin letters z_j, x_j and y_j, separated by single spaces, describing a chord. The letters are not necessarily distinct, and the same chord may appear more than once.

The last line of the test case contains the string s ($1 \le |s| \le 2 \cdot 10^5$), consisting of lowercase Latin letters.

It is guaranteed that the sum of m over all test cases does not exceed $2 \cdot 10^5$, and the sum of $|s|$ over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single integer: the minimum total effort needed to type s, or -1 if some letter of s cannot be produced at all.

Sample Input 1

```
3
1 10 100 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1
2
b a a
c b b
cab
1 1 1 -1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
0
add
-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 7 3 4
3
x y z
x z z
x y y
xxy
```

Sample Output 1

```
7
-1
15
```

Explanation of Sample 1

In the first test case, the chord "b a a" produces "b" from two presses of "a". Using two separately produced copies of "b", the chord "c b b" produces "c". Producing "cab" this way takes $\mathbf{4} + \mathbf{1} + \mathbf{2} = \mathbf{7}$ units of effort.